# The microdialysis-derived lactate–pyruvate gradient indicates anaerobic activity after brain injury

Michael S Baker,[1] Joshua M Heihre,[1] Tomasz Kuliński,[2] Claudia A Smith,[1] Berfin Barlas,[1,3] Chisomo Zimphango,[1] Ivan Timofeev,[1] Koby Baranes,[1,3] Keri L H Carpenter,[1] Mark Kotter,[1,3] Mathew R Guilfoyle,[1] Elham Rostami,[2,4] Peter J Hutchinson,[1] and Adel Helmy[1]

Author affiliations:

1 Department of Clinical Neurosciences, University of Cambridge, Cambridge Biomedical Campus, Cambridge, CB2 0QQ, UK

2 Section of Neurosurgery, Department of Medical Sciences, Uppsala University Hospital, Uppsala, SE-751 85, Sweden

3 Wellcome-MRC Cambridge Stem Cell Institute, University of Cambridge, Cambridge, CB2 0AW, UK

4 Department of Physiology and Pharmacology, Karolinska Institutet, Stockholm, SE-171 77, Sweden

Correspondence to: Michael S Baker

Division of Neurosurgery, Department of Clinical Neurosciences, University of Cambridge, Cambridge Biomedical Campus, Box 167, Cambridge, CB2 0QQ, UK

Phone: +44 1223 746453

Email: msb74@cam.ac.uk

**Running title**: Lactate–pyruvate linearity after brain injury

# Abstract

**Background**: Cerebral microdialysis (CMD) after traumatic brain injury (TBI) has focused on lactate/pyruvate ratio (LPR), overlooking lactate–pyruvate gradient (LPG, the linear regression slope of a data segment) as a proxy for anaerobic activity.

**Methods**: We retrospectively applied changepoint detection to the linear lactate–pyruvate relationship ($L = mP + b$) over time within patients from two independent CMD-monitored TBI cohorts (Cambridge, n = 510; Uppsala, n = 81), estimating per-segment LPG ($m$) and intercept ($b$, accounting for metabolic variance). *In vitro*, we tested whether rotenone-induced mitochondrial dysfunction increases LPG in hiPSC-derived neurons, as expected if LPG tracks anaerobic activity.

**Results**: In the Cambridge cohort, a median of two distinct linear segments were identified per patient (median LPG = 19.83; median $b$ = 0.54 mM). Between segments, LPR tracked LPG with heavy attenuation ($\beta$ = 0.255). LPG was negatively associated with cerebral microdialysate glucose ($p < 0.001$) and 6-month GOSE ($p = 0.039$). The Uppsala cohort confirmed pipeline robustness and replicated the LPG–glucose association ($p = 0.022$). *In vitro*, LPG was higher in rotenone-treated neurons than controls (32.9 vs. 14.2, $p < 0.001$).

**Conclusion**: Segment-level LPG provides a more accurate, outcome-associated proxy for anaerobic activity than datapoint-level LPR.



**Keywords:** cerebral metabolism; changepoint detection; mitochondrial dysfunction; neuromonitoring; neurocritical care

# 1 Introduction

Cerebral metabolic derangement is a detrimental secondary injury mechanism following traumatic brain injury (TBI).[1] Cerebral microdialysis (CMD) is an invasive technique used in neurocritical care in which a probe placed into the brain parenchyma samples extracellular fluid, enabling monitoring of local biochemistry and metabolic derangements (**Fig. 1**). The lactate/pyruvate ratio (LPR) derived from CMD has thus far served as the key clinical indicator of metabolic distress and anaerobic activity.[2] An elevated cerebral LPR is associated with poorer neurological[3–6] and brain tissue[7,8] outcomes.

However, the traditional interpretation of CMD data relies on isolated LPR measurements assessed against fixed thresholds (e.g., 25),[3,9–12] collapsing the interaction between lactate and pyruvate into a single value that obscures their linear co-variation. Here, we introduce a framework for characterising the linear lactate–pyruvate relationship to infer anaerobic activity more accurately than via datapoint-level LPR. We define anaerobic activity as increased glycolytic flux with conversion of pyruvate to lactate when oxidative metabolism is impaired (by reduced oxygen availability or delivery, or mitochondrial dysfunction), not complete tissue anoxia.

Throughout this manuscript, we use three distinct terms. (i) The lactate/pyruvate ratio (LPR) is the standard datapoint-level, unitless CMD metric of lactate divided by pyruvate at a single timepoint. (ii) The linear lactate–pyruvate relationship is estimated by fitting lactate against pyruvate ($L = mP + b$) within a segment of CMD datapoints collected over time. (iii) We term the unitless slope $m$ of this linear relationship the lactate–pyruvate gradient (LPG); intercept $b$ absorbs metabolic variance. We hypothesise that changes in LPG between segments track shifts in anaerobic activity more accurately than unadjusted, datapoint-level LPR fluctuations.

Our analyses were inspired by the findings of our past studies,[12,13] which identified a notable polarisation in the metabolic response to injury among TBI patients. These studies revealed a U-shaped distribution in the prevalence of LPR > 25, with one substantial group of patients never demonstrating this metabolic derangement and another always demonstrating LPR > 25. We anticipated this polarisation may manifest from differences in linear lactate–pyruvate relationships. Building on our group's earlier linear modelling of the lactate–pyruvate relationship,[14] with the lactate dehydrogenase (LDH) enzyme's linkage of the two substrates serving as the theoretical basis, we hypothesised that this linear relationship changes over time within patients, with distinct segments characterised by different LPG and $b$.

We pursue several interconnected objectives:

1. Develop an automated changepoint detection pipeline to retrospectively identify shifts in the CMD-derived linear lactate–pyruvate relationship between data segments within each patient.
2. Validate that segment-level linear fits outperform patient-level fits and that the intercept term $b$ is empirically necessary.
3. Characterise LPG ($m$) and $b$, including their within-patient variation.
4. Quantify how LPR is confounded by $b$ due to the hyperbolic LPR–pyruvate relationship (Equation 6).
5. Explain the previously observed polarisation in LPR > 25 prevalence by linking it to fundamental differences in linear lactate–pyruvate relationships.[12,13]
6. Assess how faithfully LPR tracks LPG and test whether LPG is associated with substrate delivery, as indicated by cerebral microdialysate glucose.
7. Test whether LPG is associated with functional outcome (6-month GOSE).
8. Assess the generalisability of our findings in an independent TBI cohort.
9. Test whether rotenone-induced mitochondrial dysfunction increases LPG in an *in vitro* neuronal model, providing mechanistic evidence that LPG reflects anaerobic activity.

This work advances the interpretation of CMD data, with LPG as a segment-level alternative to datapoint-level LPR. Through identifying LPG as a more direct and outcome-associated indicator of anaerobic activity than the LPR, confirming findings in an independent cohort, and providing *in vitro* evidence, we provide a novel framework for understanding inter- and intra-patient metabolic variability and the potential to enhance retrospective analysis for guiding personalised therapeutic interventions in TBI.

# 2 Materials and methods

## 2.1 Study population

This retrospective analysis utilised a comprehensive neuromonitoring dataset from 619 patients with acute TBI, collected at the neurocritical care unit of Addenbrooke's Hospital, Cambridge, UK, between December 1997 and January 2017. The head injury programme, under which the data was collected, received approval from the Cambridge Local Research Ethics Committee (REC reference: 97/290; IRAS project ID: 39184) on December 5, 1997, and from

Addenbrooke's Hospital's Research and Development Department. Informed consent was obtained from the next of kin or a legally authorised representative, in written or verbal form, with provision for deferred consent. The cohort comprised patients aged 16 and over, with a median age of 37 years (IQR 24–52), and was predominantly male (76.3%). Patients were managed according to standardised neurocritical care protocols and monitored using established multimodal monitoring techniques; for complete details on clinical management protocols and multimodal monitoring specifications, refer to Supplementary materials SM1. Previous analyses on subsets and the entirety of this cohort have been published.[3,6,12]

We also analysed data from an independent cohort of 81 CMD-monitored severe TBI patients (median age 39 years, IQR 20–56; 75.3% male) admitted to Uppsala University Hospital, Sweden, between 2008 and 2020 (ethical approval: Swedish Ethical Review Authority, Uppsala, Sweden; approval numbers: #2010/138, #2010/138/1, #2010/379, #2015/224, #2020-05462; written informed consent from next-of-kin). Details of this cohort, including study population and clinical management, are provided in Supplementary materials SM3.

## 2.2 Data integration and pre-processing

The present study draws on the integrated, pre-processed multimodal dataset that our group generated for earlier studies.[3,6] The raw data, 56,307 observations across 619 patients, were imported using custom R scripts. The resulting dataset, comprising 56,250 multimodal observations from 616 patients, was accessed for the current study on May 1, 2025, after all identifying information had been anonymised. The dataset was then subjected to additional pre-processing for linear lactate–pyruvate dynamics. The pre-processing pipeline produced a cleaned dataset containing 50,797 total observations (with complete data for lactate, pyruvate, and LPR) across 510 patients. In the analysed Cambridge and Uppsala cohorts, CMD lactate and pyruvate were sampled at approximately hourly intervals; the CMD data included in the analysis spanned a median of 5.2 days per patient (Cambridge IQR 3.4–8.0; Uppsala IQR 3.4–7.4; Results section 3.1, Tables S1 and S4). Full pre-processing details may be found in the Supplementary materials SM2 for the Cambridge cohort and SM3.iii for the Uppsala cohort.

## 2.3 Linear characterisation of lactate–pyruvate data

### 2.3.i Theoretical basis

The intracellular LDH reaction is the foundation of our framework centred on the linear lactate–pyruvate relationship. This reaction is fast and reversible, operating in near-equilibrium within the living cell:

$$Lactate + NAD^{+} \overset{\text{LDH}}{\longleftrightarrow} Pyruvate + NADH + H^{+} \qquad (1)$$

At near-equilibrium, the enzyme maintains the products/reactants ratio (the mass-action ratio) very close to the reaction's thermodynamic equilibrium constant ($K_{eq}$). Thus, the relationship is described by the law of mass action:

$$K_{eq} \approx \frac{[Pyruvate][NADH][H^{+}]}{[Lactate][NAD^{+}]} \qquad (2)$$

We can rearrange this equation to define the relationship between the CMD-measured metabolites, lactate (L) and pyruvate (P):

$$L \approx \left(\frac{1}{K_{eq}} \times [H^{+}] \times \frac{[NADH]}{[NAD^{+}]}\right) \times P \qquad (3)$$

The entire parenthetical term in Equation 3 is a composite parameter that we denote *m* and refer to as the lactate–pyruvate gradient (LPG):

$$L = mP \qquad (4)$$

LPG ($m$) remains stable for a segment of data when the product $\left(\frac{[NADH]}{[NAD^{+}]} \times [H^{+}]\right)$ is stable; this does not require that the redox state or proton concentration be individually constant. Equation 4 represents a simplified model relative to Equation 5. An empirical model with intercept *b* can be fit to account for remaining metabolic factors, such as those arising from non-LDH-linked substrate sources or sinks:

$$L = mP + b \qquad (5)$$

Thus, the intercept term *b* absorbs non-LDH-linked contributions to lactate and pyruvate, enabling LPG ($m$) to isolate the LDH-driven signal reflecting anaerobic activity.

Crucially, when $b \neq 0$, dividing Equation 5 by pyruvate reveals a hyperbolic relationship between LPR and pyruvate (see Equation 6) in which LPR is displaced from LPG ($m$) and varies with substrate concentration even when LPG reflects a stable underlying metabolic signal.

$$LPR = \frac{L}{P} = m + \frac{b}{P} \qquad (6)$$

**Fig. S1** illustrates this transformation conceptually, demonstrating how the sign and magnitude of $b$ determine the direction and degree of curvature of the resulting LPR–pyruvate hyperbola, while the LPG ($m$) sets the horizontal asymptote toward which the LPR converges at high pyruvate concentrations.

Our framework operates on two assumptions: that the LDH reaction is always operating in a state of near-equilibrium,[15] and that extracellular CMD measurements serve as a valid proxy for these intracellular dynamics. Monocarboxylate transporters (MCT1, MCT2, and MCT4) on neurons and glial cells facilitate rapid metabolic coupling of lactate and pyruvate between intracellular and extracellular compartments,[16] enabling extracellular CMD to reflect local intracellular LDH kinetics at the probe tip. We use this framework to track LPG ($m$), a direct indicator of anaerobic activity inferred via CMD, between data segments over time.

## 2.3.ii Segmentation pipeline

To identify temporally distinct linear segments within each patient's lactate–pyruvate data (Equation 5), we developed an automated segmentation pipeline using changepoint detection. The pipeline employs the Pruned Exact Linear Time (PELT) algorithm[17,18] to detect changepoints in the linear lactate–pyruvate relationship over time, with a minimum segment length of 6 datapoints. The optimal number of segments was then selected using the Kneedle algorithm[19,20] to identify the elbow point on the cost-complexity curve. Segments partition each patient's CMD data series into temporally ordered, non-overlapping blocks of datapoints separated at boundaries by one sampling interval (~1 h). Changepoints are introduced when the data indicate a metabolic transition large enough to warrant a fitted shift in the linear lactate–pyruvate relationship, rather than pinpointing the instantaneous time of metabolic change. Ordinary least squares (OLS) regression was then fit within each resulting segment to obtain segment-level LPG ($m$) and intercept ($b$) values. For downstream analyses, we retained only segments with positive LPG ($m > 0$), consistent with $m > 0$ as implied by Equation 3, where $m$ comprises only positive terms under near-equilibrium LDH kinetics. Complete pipeline details and code are provided in the published repository.

## 2.4 *In vitro* model

To independently test whether LPG is sensitive to anaerobic activity, we induced mitochondrial dysfunction in a reductionist *in vitro* hiPSC-derived neuronal model. Glutamatergic neurons were generated via optimised inducible overexpression (opti-ox) of neurogenin 2 (NGN2) in hiPSCs and maintained at 37°C in a humidified 5% $CO_2$ atmosphere (full protocol in Supplementary materials SM4). On Day 21 post-induction, neurons underwent a complete media change to a test medium supplemented with 5 mM glucose without exogenous lactate, pyruvate, or L-glutamine, containing either rotenone (10 nM, ≥ 95% purity; Sigma-Aldrich, St. Louis, MO, USA; a complex I inhibitor of the mitochondrial electron transport chain [ETC], dissolved in DMSO) or vehicle control. Samples were collected from parallel wells at timepoints between 1 and 24 hours post-treatment and analysed on ISCUSflex analysers (M Dialysis AB, Stockholm, Sweden), mirroring our clinical workflow.

## 2.5 Statistical analysis and code availability

Multiple statistical approaches were used, including OLS regression, Fisher's z-transformed correlation comparisons, model comparison via Akaike Information Criterion (AIC), linear mixed-effects (LME) models with nested random effects, nonparametric tests for group comparisons with Benjamini-Hochberg correction for multiple testing, linear contrasts via estimated marginal means (EMMs), and analysis of covariance (ANCOVA) for comparing *in vitro* regression lines. EMMs represent the model-estimated typical values within each group, adjusted for the model's random effects and weighting structure. All analyses were performed using R (v4.5.2; https://www.r-project.org/) and Python (v3.9.6; https://www.python.org/). The complete, annotated source code of this project is publicly available at https://github.com/msb-jr/linear-LP-gradient. The relation of each script to each numbered section of the manuscript and Supplementary materials is specified in the repository. Where directional hypotheses were pre-specified based on theoretical expectation or established literature (e.g., higher LPR associated with worse outcomes), one-sided tests were used; all other tests were two-sided. The directionality of each test is specified in the published code. The Uppsala cohort was analysed using the same published pipeline and scripts; all statistical methods were identical to those applied to the Cambridge cohort.

# 3 Results

Sections 3.1–3.6 describe the Cambridge cohort; results from the independent Uppsala cohort (3.7) and *in vitro* validation (3.8) follow.

## 3.1 Segmentation pipeline output

The lactate–pyruvate linear segmentation pipeline was applied to 50,797 datapoints across 510 patients (median 88 datapoints per patient, IQR 56–135, spanning a median of 5.2 days, IQR 3.4–8.0 days). The pipeline identified 1,465 data segments in total. After retaining only segments with positive LPG ($m > 0$), 1,339 segments (91.4%) comprising 48,504 datapoints (95.5%) were retained across 506 patients (99.2%).

Retained segments contained a median of 30 datapoints (IQR 20–42) collected over a median span of 41.4 hours (IQR 28.3–60.4 h). Each retained patient had a median of 2 retained segments (IQR 2–3; range 1–7). Segmentation details are reported in **Table S1**. **Fig. 2** visualises segmentation results for representative patients. **Fig. S2** illustrates the complete pipeline process for a patient in whom one segment was discarded due to a non-positive LPG.

## 3.2 Validation of the linear framework

We validated two core assumptions of our framework. First, we tested whether fitting multiple linear segments per patient captures the lactate–pyruvate relationship more accurately than fitting a single regression line to each patient's entire monitoring period. A weighted LME model on Fisher's z-transformed Pearson correlations (1,465 segments across 510 patients) confirmed that segment-level correlations (EMM $r = 0.83$, 95% CI: 0.82–0.85) were significantly higher than patient-level correlations (EMM $r = 0.75$, 95% CI: 0.73–0.77; $p < 0.001$), indicating that the positive linear lactate–pyruvate relationship is more clearly resolved when patients' data is segmented and justifying the use of changepoint-based segmentation. All subsequent analyses were performed exclusively on retained segments ($m > 0$).

Second, we tested whether including a non-zero intercept ($b$) improves segment-level linear fits compared to an origin-constrained model ($\mathrm{L} = m\mathrm{P}$, Equation 4). Across 1,339 retained segments from 506 patients, the typical improvement was substantial ($\Delta\mathrm{AIC} = 10.45$, 95% CI: 9.46–11.45; $p < 0.001$), with 57.2% of individual segments showing a clear preference for the unconstrained model ($\Delta\mathrm{AIC} > 2$). These results empirically confirm that the two-parameter model (L = $m$P + $b$, Equation 5) better describes the observed lactate–pyruvate data, supporting the inclusion of the intercept to isolate the LPG as a proxy for anaerobic activity.

## 3.3 Linear parameters and the hyperbolic consequence

### 3.3.i Characterisation

Across the 1,339 retained segments of 506 patients (**Fig. 3A–B**), the cohort median LPG was 19.83 (IQR: 15.48–25.10) and the median intercept ($b$) was 0.54 mM (IQR: 0.13–1.12 mM), with a median within-segment $R^2$ of 0.69 (IQR: 0.54–0.80). Among the 466 patients with two or more retained segments, within-patient variation was substantial: the median intra-patient range of LPG was 12.28 (IQR: 6.33–21.20) and for $b$ was 1.69 mM (IQR: 0.77–3.00 mM).

Stratification by the sign of the intercept revealed two distinct patterns (**Fig. 3C–H**). Segments with a positive intercept (Type $P_b$; 967 [72.2%] segments, 476 patients; median $b$ = 0.87 mM; **Fig. 3C–D**) had lower LPG (median LPG = 17.97, IQR: 13.33–22.03) and moderate linear fits (median $R^2$ = 0.63, IQR: 0.48–0.77), producing decreasing LPR hyperbolae as pyruvate increased (**Fig. 3D**). Segments with a negative intercept (Type $N_b$; 372 [27.8%] segments, 276 patients; median $b$ = -0.55 mM; **Fig. 3E–F**) had markedly higher LPG (median LPG = 29.04, IQR: 23.83–35.72) and stronger linear fits (median $R^2$ = 0.85, IQR: 0.74–0.91), producing increasing LPR hyperbolae toward the asymptote equal to the LPG ($m$) (**Fig. 3F**). These empirical patterns mirror the conceptual framework illustrated in **Fig. S1**.

### 3.3.ii Within-segment LPR error

The presence of a non-zero intercept causes the LPR to deviate from the underlying LPG in a substrate-dependent manner (**Table S2**). This deviation is most pronounced at low pyruvate concentrations: at the segment-minimum pyruvate (median 64 μM, IQR: 41–89 μM), the overall median relative LPR error was 66.8% (IQR: 20.2%–190.6%). In Type $P_b$ segments, the LPR exceeded the LPG by a median of 103.4% (IQR: 39.1%–254.7%) at minimum pyruvate, while in Type $N_b$ segments, the LPR underestimated the LPG by a median of -19.9% (IQR: -38.0% to -8.6%). Even at higher substrate levels (segment-maximum pyruvate, median 172 μM, IQR: 139–225 μM), the median LPR error remained 23.3% (IQR: 7.1%–55.6%). Furthermore, LPR varied substantially within segments despite a stable LPG: the median within-segment LPR range was 14.4 (IQR: 9.2–27.6), driven entirely by substrate-dependent fluctuations along the hyperbolic curve. These findings demonstrate that the LPR systematically fails to track the level of anaerobic activity reflected by the LPG, with the direction and magnitude of the deviation determined by the sign and magnitude of the intercept *b* and the prevailing pyruvate concentration. **Fig. 4** demonstrates this mismatch in one patient's time series, where the LPR remained in a similar range despite a near-threefold decrease in the LPG between segments.

## 3.4 Explaining metabolic polarisation

To explain the previously observed polarisation of LPR values around the clinical threshold of 25,[12,13] we compared the linear parameters of two cohorts: patients whose LPR always exceeded 25 ("Always LPR > 25"; 39 patients, 82 retained segments) and those whose LPR never exceeded 25 ("Never LPR > 25"; 38 patients, 92 retained segments). Notably, both cohorts had similarly negligible proportions of datapoints in discarded segments ($m \leq 0$) prior to filtering (median 0.0% in both), supporting the comparability of the two groups on the basis of retained data.

The "Always LPR > 25" cohort had significantly higher LPG (median LPG = 27.39, IQR: 23.82–33.78) compared to the "Never LPR > 25" cohort (median LPG = 14.59, IQR: 12.44–18.20; $p < 0.001$), significantly higher intercepts (median $b$ = 1.62 mM, IQR: 0.58–2.76 vs. 0.19 mM, IQR: -0.15–0.37; $p < 0.001$), and a significantly higher proportion of Type $P_b$ segments (median 100.0% vs. 66.3%; $p < 0.001$) (**Table S3**). These differences in underlying linear parameters produce fundamentally different hyperbolic LPR–pyruvate curves: the combination of higher LPG and higher positive *b* in the "Always" cohort generates LPR values that are generally above 25 across the physiological pyruvate range, while the lower LPG and near-zero *b* in the "Never" cohort produce flatter or inverted curves that keep LPR below the threshold. Thus, the previously unexplained polarisation of LPR about 25 is a direct mathematical consequence of differences in the underlying linear lactate–pyruvate relationships.

## 3.5 Substrate delivery and LPR fidelity

Next, we investigated the relationships between cerebral glucose (a proxy for substrate delivery), LPR, and the linear parameters LPG (*m*) and *b* (**Fig. 5A–D**). Glucose has a dual association with the system: it tracks segment-level anaerobic activity (LPG), and it simultaneously drives pyruvate supply, generating artefactual LPR fluctuations through the hyperbolic relationship (Equation 6). At the level of individual observations, glucose was strongly and positively associated with pyruvate ($\beta = 13.71$ μM/mM, $p < 0.001$; residual SD = 30.8 μM), supporting that glucose drives pyruvate and validating the examination of hyperbolic LPR–glucose associations.

Within segments of stable LPG, the LPR exhibited a hyperbolic association with glucose that depended on the sign of the intercept $b$ (**Fig. 5A–B**). In Type $P_b$ segments, increasing glucose was associated with decreasing LPR ($p < 0.001$; **Fig. 5A**), while in Type $N_b$ segments the association reversed ($p < 0.001$; **Fig. 5B**; $p$-values Benjamini-Hochberg corrected, n = 2).

At the between-segment level, the LPR correlated positively with LPG ($p < 0.001$; **Fig. 5C**), but the association was heavily attenuated ($\beta = 0.255$, i.e., 0.255-unit increase in segment median LPR per unit increase in LPG, compared to a theoretical $\beta = 1$ if LPR perfectly tracked LPG). Part of the attenuation is explained by a significant negative coupling between LPG and $b$ ($\beta = -0.097$ mM per unit LPG, $p < 0.001$). Given Equation 6 (LPR = $m + b/P$): when LPG ($m$) increases and $b$ simultaneously decreases, the opposing change in $b$ partially offsets the effect of higher LPG on the LPR.

Finally, the LPG showed a robust negative association with segment-level median glucose ($\beta = -1.75$ mM$^{-1}$, $p < 0.001$; **Fig. 5D**), consistent with the expected physiological response: reduced substrate delivery is associated with increased anaerobic activity. The segment-level median LPR was also negatively associated with glucose ($\beta = -2.41$ mM$^{-1}$, $p < 0.001$; **Fig. 5D**), but with a coefficient larger in magnitude than that of LPG. This disproportionately large response is consistent with the summation of two effects: (i) the true physiological signal (lower LPG at higher glucose) and (ii) a Type $P_b$ (72.2% of segments) substrate-dependent artefact (higher glucose → higher pyruvate → deflated $^{+}b/P$ term → further LPR deflation). Thus, while both LPG and LPR decrease with increasing glucose, LPR's response conflates the anaerobic signal with the intercept-dependent artefact, whereas LPG isolates the physiological relationship.

## 3.6 Outcome association

Additionally, we tested whether the linear parameters LPG ($m$) and $b$ are associated with 6-month functional outcome using the Glasgow Outcome Scale-Extended (GOSE), summarised into four categories: Death (GOSE 1; 87 patients, 217 segments), Severe disability (GOSE 2–

4; 117 patients, 310 segments), Moderate disability (GOSE 5–6; 160 patients, 423 segments), and Good recovery (GOSE 7–8; 112 patients, 305 segments).

The EMMs for each outcome group are shown in **Fig. 6A–C**. Validating established literature,[3,10] LPR was significantly associated with outcome (linear contrast $p = 0.030$; **Fig. 6A**). However, the LPR–outcome trend was largely driven by the Death group (EMM = 30.2, 95% CI: 27.6–32.8), with the Severe, Moderate, and Good groups clustering closely together (EMMs: 26.0, 26.2, 26.7).

Crucially, higher values of LPG were also significantly associated with worse outcome (linear contrast $p = 0.039$; **Fig. 6B**), but unlike the LPR, the LPG showed a consistent monotonic trend across all four outcome categories, decreasing from 22.1 (95% CI: 20.6–23.6) in the Death group to 20.3 (95% CI: 19.0–21.5) in the Good recovery group. This graded relationship supports the clinical relevance of LPG: segments belonging to patients with worse neurological outcomes were characterised by a steeper linear lactate–pyruvate relationship (higher LPG), indicating higher levels of anaerobic cerebral metabolism.

The intercept $b$ showed no significant association with outcome (linear contrast $p = 0.806$; **Fig. 6C**), consistent with its role as an empirical adjustment parameter rather than a direct indicator of the LDH-linked metabolic signal.

## 3.7 Independent Uppsala cohort

To assess generalisability, an independent centre (Uppsala University Hospital, Uppsala, Sweden) applied the segmentation pipeline and all downstream analyses to a TBI cohort of 81 patients (8,150 CMD observations; Supplementary materials SM3). The pipeline identified 232 segments, of which 215 (92.7%) were retained ($m > 0$), encompassing 7,781 datapoints (95.5%) across all 81 patients (100%). Each patient had a median of 2 retained segments (IQR 2–3; **Table S4**; **Fig. S3**).

Segment-level Pearson correlations (EMM $r = 0.83$) were significantly higher than patient-level correlations (EMM $r = 0.80$; $p = 0.015$), replicating the Cambridge finding and confirming that segmentation improves the resolution of the linear lactate–pyruvate relationship. The intercept model ($L = mP + b$) significantly outperformed the origin-constrained model (typical $\Delta AIC = 7.00$, 95% CI: 4.85–9.15, $p < 0.001$), with 42.8% of segments showing a clear preference for the unconstrained model ($\Delta AIC > 2$).

Across the retained segments, the median LPG (*m*) was 20.16 (IQR: 17.16–24.93), median intercept (*b*) was 0.12 mM (IQR: -0.15–0.76), and median within-segment $R^2$ was 0.66 (IQR: 0.53–0.78). Among the 76 patients with two or more retained segments, the median intra-patient range of LPG was 11.36 (IQR: 5.91–18.23). LPR varied substantially within segments despite stable LPG (median within-segment LPR range: 10.4, IQR: 7.4–14.3; **Table S5**).

The within-segment hyperbolic LPR–glucose association was significant in Type $P_b$ segments ($p < 0.001$; **Fig. 5E**) but did not reach significance in Type $N_b$ segments ($p = 0.090$; **Fig. 5F**). Glucose was positively associated with pyruvate (β = 16.38 μM/mM, $p < 0.001$; residual SD = 28.3 μM). The segment-level association between LPR and LPG (β = 0.140, $p = 0.005$; **Fig. 5G**) replicated the direction of the Cambridge finding (β = 0.255), with both falling substantially short of the theoretical β = 1 if LPR perfectly tracked LPG. The heavier attenuation of the LPR–LPG association in the Uppsala cohort is in line with the more negative coefficient of the LPG–*b* relationship (β = -0.142 mM per unit LPG, $p < 0.001$). The LPG was negatively associated with segment-level median glucose (β = -1.20 $mM^{-1}$, $p = 0.022$), while segment-level median LPR demonstrated a more extreme trend with glucose (β = -3.95 $mM^{-1}$, $p < 0.001$; **Fig. 5H**) consistent with amplification from an additional substrate-dependent artefact associated with the majority of segments' Type $P_b$ identity (61.4%).

In the Uppsala cohort (69 patients with outcome data: Death, 6; Severe, 29; Moderate, 16; Good, 18), neither LPR (linear contrast $p = 0.470$) nor the LPG (linear contrast $p = 0.784$) were significantly associated with 6-month GOSE. The intercept *b* also showed no significant association (linear contrast $p = 0.546$; **Fig. 6D–F**).

## 3.8 *In vitro* validation

To test whether LPG (*m)* is sensitive to anaerobic activity, we induced mitochondrial dysfunction in hiPSC-derived neurons with rotenone (10 nM), a complex I inhibitor, and examined the linear lactate–pyruvate relationship (**Fig. 7**). Under control conditions, neurons exhibited an LPG of 14.2 ($R^2 = 0.66$, n = 34). Rotenone treatment more than doubled LPG to 32.9 ($R^2 = 0.80$, n = 34), with the difference confirmed by ANCOVA (LPG shift $p < 0.001$; residual df = 64). The intercept (*b*) was near zero in both conditions (control: 0.026 mM; rotenone: -0.107 mM), though the shift was also significant ($p < 0.001$). These results provide direct evidence that inhibition of the mitochondrial ETC increases the LPG, consistent with the interpretation of LPG as an indicator of anaerobic activity.

# 4 Discussion

This study introduces a framework for isolating the lactate–pyruvate gradient (LPG, $m$) estimated by linear regression (Equation 5, $L = mP + b$) as a more direct and stable indicator of anaerobic activity than the LPR. To our knowledge, this is the first study to systematically characterise the linear lactate–pyruvate relationship as monitored by CMD and to track it over time within individual patients. We demonstrate that the LPG is negatively associated with substrate delivery, significantly associated with functional outcome, and mechanistically linked to mitochondrial dysfunction, and we replicated core empirical findings in the Uppsala cohort. In contrast, the LPR, although also associated with outcome, is systematically displaced from the LPG by the linear intercept ($b$), which captures non-LDH-linked metabolic variance. Thus, the LPR is a less reliable marker of anaerobic activity.

The hypothesis that LDH-driven lactate–pyruvate dynamics are detectable via CMD and linear modelling was strongly supported by our data: 91.4% of detected linear segments had positive LPG, consistent with $m > 0$ as implied by Equation 3, where $m$ comprises only positive terms under near-equilibrium LDH kinetics. These retained segments had a median duration of 41.4 hours. The minority of segments with non-positive LPG (8.6%) may reflect Simpson's paradox[21] arising from insufficient temporal resolution, or biological complexity such as non-LDH-linked lactate sources (e.g., cell death) (**Fig. S2**). The high retention rate indicates that the linear lactate–pyruvate relationship is a robust and prevalent feature of CMD data in TBI.

A key methodological contribution of this study was the use of data-driven, automated changepoint detection to identify temporally distinct linear trends within each patient's monitoring data. This approach was empirically validated: segment-level correlations significantly exceeded patient-level correlations, confirming that patients experience multiple distinct linear relationships throughout their monitoring rather than a single static trend. The substantial within-patient variation in LPG (median range 12.28) further underscores that the level of anaerobic activity is not fixed but shifts over time, potentially reflecting changes in substrate delivery, mitochondrial function, or therapeutic intervention.

The empirical necessity of the intercept ($b$) was confirmed by AIC model comparison. While $b$ is essential for accurately isolating LPG, its presence destabilises the LPR: given Equation 6 ($LPR = m + b/P$), LPR deviates from LPG ($m$) in a substrate-dependent manner amplified at low pyruvate. This error was striking: in positive-intercept segments, median LPR error at minimum pyruvate was +103.4%, and within a single segment of stable LPG the LPR varied by a median of 14.4 units, driven entirely by substrate fluctuations. Clinically, bedside changes in LPR may reflect pyruvate supply rather than genuine shifts in anaerobic metabolism.

The substrate delivery and LPR fidelity analyses provided converging evidence that LPR poorly reflects underlying LPG. Within segments of stable LPG, LPR fluctuations largely reflect changes in glucose and pyruvate supply (with possible hemodynamic contributions from intracranial pressure and cerebral perfusion pressure) and correspond to movement along the fitted linear lactate–pyruvate relationship. Accordingly, glucose drove LPR in opposite directions depending on the sign of $b$, illustrating the artefactual nature of these fluctuations. Between segments, LPR was a heavily attenuated proxy for LPG, partly because of a significant negative LPG–$b$ coupling: as the fitted line steepens (higher LPG), $b$ tends to fall, and given Equation 6 ($LPR = m + b/P$), this opposing change partially cancels the effect of higher LPG. Although LPG was robustly and negatively associated with glucose, LPR showed a stronger negative association, reflecting summation of the true physiological signal with the substrate-dependent artefact in the majority Type $P_b$ segments.

The clinical relevance of the LPG was established by its significant association with 6-month GOSE. Higher LPG was associated with worse outcome across a graded decline from Good recovery to Death. In contrast, the LPR–outcome association was driven primarily by the mortality group, with surviving outcome categories clustering together. The linear intercept ($b$) showed no association with outcome, in line with its role as an empirical adjustment parameter. Together, these findings position the LPG as an outcome-associated, physiologically grounded metric that the LPR only imperfectly approximates.

In the independent Uppsala cohort, several key findings were replicated: comparable segmentation output (92.7% retention, median $R^2 = 0.66$), near-identical within-patient LPG variation (median range 11.36 vs. 12.28), attenuated LPR tracking of LPG ($\beta = 0.140$ vs. 0.255, consistent with a stronger negative LPG–$b$ coupling), and replicated negative LPG–glucose association ($\beta = -1.20$, $p = 0.022$). The within-segment LPR–glucose association was only significant in Type $P_b$ segments and did not reach significance in Type $N_b$ segments ($p = 0.090$), possibly due to the smaller sample of $N_b$ segments. The outcome association did not reach significance, most likely reflecting the smaller sample (69 vs. 476 patients with outcome data) and limited power. Replication of the structural, physiological, and methodological findings across two centres nonetheless strengthens confidence in generalisability.

The *in vitro* experiment tested whether LPG increases when anaerobic activity is driven by mitochondrial dysfunction. Rotenone inhibits complex I of the electron transport chain, impairing oxidative phosphorylation and shifting cellular metabolism toward glycolysis; in hiPSC-derived neurons, this more than doubled LPG (from 14.2 to 32.9), supporting our interpretation of LPG as an indicator of anaerobic activity. Notably, the intercept $b$ remained near zero in both conditions, suggesting that in a controlled cellular environment without the complexity of the *in vivo* extracellular milieu, the simplified model ($L = mP$, Equation 4) is a close approximation. Together with the *in vivo* glucose association, these findings provide complementary evidence that LPG captures the metabolic response to two of the primary drivers of secondary injury in TBI: substrate limitation and mitochondrial dysfunction.

This study has inherent limitations. First, our framework is retrospective: identifying linear segments requires sufficient data over time, and the accuracy of segment-level parameters depends on the duration and stability of the underlying trend. Real-time elucidation of linear trends at the bedside currently lacks feasibility with standard hourly CMD sampling. Higher-frequency or continuous monitoring may enable online LPG tracking in the future, but purpose-built algorithms for real-time changepoint detection remain to be developed. Complementary non-invasive modalities, such as hyperpolarized [1-$^{13}$C]pyruvate magnetic resonance spectroscopy for real-time cerebral metabolic flux imaging,[22] address overlapping metabolic questions and were not evaluated in the present CMD-focused analysis. Second, our analysis focused exclusively on segments with positive LPG ($m > 0$); while this decision was theoretically motivated by the LDH equilibrium and segments with non-positive LPG (8.6% of all segments) are likely artefactual (as discussed above), this filtering means a small proportion of monitoring time is excluded from downstream analyses. Third, the reductionist *in vitro* model, while providing evidence for the LPG–mitochondrial dysfunction link, does not wholly replicate *in vivo* cytoarchitecture and lacks vascular and blood-brain barrier components; it employed a single cell type and inhibitor, and further studies using alternative models of anaerobic activity would strengthen these findings. Finally, while the Uppsala cohort replicated most findings, the outcome association was not replicated, and further validation in larger cohorts is warranted.

In conclusion, this study identifies the lactate–pyruvate gradient (LPG) as a novel CMD-derived metric that provides a stable readout of anaerobic cerebral metabolism over a given data segment; it is not subject to the substrate-dependent confounding that renders the LPR an unreliable proxy. The linear lactate–pyruvate relationship is validated across two independent clinical cohorts, outcome-associated, and linked to mitochondrial dysfunction through *in vitro* modelling. While currently best suited for retrospective analysis, this framework establishes a foundation for developing more precise, individualised approaches to monitoring and understanding cerebral metabolism after traumatic brain injury.


# Acknowledgements

We thank the patients at Addenbrooke's Hospital and Uppsala University Hospital who took part in this research.

The authors acknowledge Professor Mike Murphy, FMedSci (MRC Mitochondrial Biology Unit, University of Cambridge) for his input and correspondence concerning the metabolic considerations of our proposed framework.

MSB and JMH are supported by Gates Cambridge Scholarships from the Gates Cambridge Trust. KLHC is supported by awards from NIHR Invention for Innovation (i4i), the Wellcome Trust Developing Concept Fund (DCF), the Rosetrees Trust, and the Addenbrooke's Charitable Trust, Cambridge University Q14 Hospitals. ER is supported by Region Uppsala, Uppsala University Hospital, the Kjell och Märta Beijers stiftelse (Kjell and Märta Beijer Foundation), and the Swedish Research Council (project 2024-02725). PJH is supported by the NIHR (Senior Investigator Award, NIHR Global Health Research Group on Acquired Brain and Spine Injury, NIHR Health Tech Research Centre for Brain Injury, Cambridge Biomedical Research Centre) and the Royal College of Surgeons of England. PJH and AH are supported by TBI Reporter (MRC, NIHR, Ministry of Defence, Alzheimer's Research UK). AH is supported by the Medical Research Council/Royal College of Surgeons of England Clinical Research Training Fellowship (Grant no. G0802251), MRC Grants (MR/X021882/1, MR/Y008502/1, MR/Z504890/1), the NIHR Biomedical Research Centre and the NIHR Health Technology Research Centre and NIHR Grant (NIHR129748). The views expressed are those of the authors and not necessarily those of the NIHR, the Department of Health and Social Care, or other supporting entities. The supporting entities had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript.

Under author direction, large language models were used to assist in code and manuscript drafting. The authors reviewed all outputs and assume full responsibility for the work.

# Author contributions

Conceptualization: MSB, MRG. Data curation: MSB, JMH, TK, IT, MRG. Formal analysis: MSB. Investigation: MSB, JMH. Methodology: MSB, JMH. Project administration: MSB, JMH, AH. Resources: BB, IT, MK, MRG, ER, PJH, AH. Software: MSB. Supervision: CAS, CZ, KB, KLHC, MK, ER, PJH, AH. Validation: JMH, TK, ER. Visualization: MSB, JMH. Writing – original draft: MSB. Writing – review & editing: all authors.

# Statements and declarations

## Ethical considerations

This study was conducted in accordance with the Declaration of Helsinki. For the Cambridge cohort, the head injury research programme was approved by the Cambridge Local Research Ethics Committee (REC reference: 97/290; IRAS project ID: 39184) on December 5, 1997. Addenbrooke's Hospital's Research and Development Department also approved the study. For the Uppsala cohort, ethical approval was granted by the Uppsala County branch of the Swedish Ethical Review Authority (approval numbers: #2010/138, #2010/138/1, #2010/379, #2015/224, and #2020-05462). The *in vitro* component used human induced pluripotent stem cell-derived neurons and did not involve human participants; separate ethical approval was not required.

## Consent to participate

For the Cambridge cohort, informed consent was obtained from the next of kin or a legally authorised representative, in written or verbal form, with provision for deferred consent. For the Uppsala cohort, written informed consent was obtained from next-of-kin.

## Consent for publication

Not applicable. No individually identifiable patient data are presented; all data were anonymised.

## Declaration of conflicting interest

The authors declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

## Funding statement

The authors received no financial support for the research, authorship, and/or publication of this article.

## Data availability

Cambridge cohort: Consent obtained at the time of data collection (1997–2017) did not include provisions for unrestricted public data sharing. De-identified data are available to qualified researchers for ethically approved studies upon reasonable request, subject to a data use agreement. Requests should be directed to the Cambridge Neurochemistry Research Group, Division of Neurosurgery (enquiries.neurochem@wbic.cam.ac.uk).

Uppsala cohort: De-identified data are available upon reasonable request. Requests should be directed to Elham Rostami (elham.rostami@uu.se).

*In vitro* data: Available upon reasonable request. Requests should be directed to Joshua M Heihre (jmh272@cam.ac.uk).

# Supplementary material

Supplementary material is available online.

# References


1. Demers-Marcil S and Coles JP. Cerebral metabolic derangements following traumatic brain injury. *Curr Opin Anaesthesiol* 2022; 35: 562–569.

2. Thelin EP, Carpenter KL, Hutchinson PJ, et al. Microdialysis monitoring in clinical traumatic brain injury and its role in neuroprotective drug development. *AAPS* J 2017; 19: 367–376.

3. Timofeev I, Carpenter KL, Nortje J, et al. Cerebral extracellular chemistry and outcome following traumatic brain injury: A microdialysis study of 223 patients. *Brain* 2011; 134: 484–494.

4. Paraforou T, Paterakis K, Fountas K, et al. Cerebral perfusion pressure, microdialysis biochemistry and clinical outcome in patients with traumatic brain injury. *BMC Res Notes* 2011; 4: 540.

5. Stein NR, McArthur DL, Etchepare M, et al. Early cerebral metabolic crisis after TBI influences outcome despite adequate hemodynamic resuscitation. *Neurocrit Care* 2012; 17: 49–57.

6. Guilfoyle MR, Helmy A, Donnelly J, et al. Characterising the dynamics of cerebral metabolic dysfunction following traumatic brain injury: A microdialysis study in 619 patients. *PLoS One* 2021; 16: e0260291.

7. Marcoux J, McArthur DA, Miller C, et al. Persistent metabolic crisis as measured by elevated cerebral microdialysis lactate-pyruvate ratio predicts chronic frontal lobe brain atrophy after traumatic brain injury. *Crit Care Med* 2008; 36: 2871–2877.

8. Bernini A, Magnoni S, Miroz JP, et al. Cerebral metabolic dysfunction at the acute phase of traumatic brain injury correlates with long-term tissue loss. *J Neurotrauma* 2023; 40: 472–481.

9. Bellander BM, Cantais E, Enblad P, et al. Consensus meeting on microdialysis in neurointensive care. *Intensive Care Med* 2004; 30: 2166–2169.

10. Hutchinson PJ, Jalloh I, Helmy A, et al. Consensus statement from the 2014 International Microdialysis Forum. *Intensive Care Med* 2015; 41: 1517–1528.

11. Venturini S, Bhatti F, Timofeev I, et al. Microdialysis-based classifications of abnormal metabolic states after traumatic brain injury: A systematic review of the literature. *J Neurotrauma* 2023; 40: 195–209.

12. Baker MS, Venturini S, Lindblad C, et al. Defining metabolic abnormalities in acute human traumatic brain injury with cerebral microdialysis and multimodality monitoring. *PLoS One* 2025; 20: e0331310.

13. Khellaf A, Garcia NM, Tajsic T, et al. Focally administered succinate improves cerebral metabolism in traumatic brain injury patients with mitochondrial dysfunction. *J Cereb Blood Flow Metab* 2022; 42: 39–55.

14. Guilfoyle MR. *Contusion progression in traumatic brain injury*. PhD Thesis, University of Cambridge, UK, 2019.

15. Nelson DL, Cox MM and Hoskins AA. *Lehninger principles of biochemistry*. 8th ed. New York: Macmillan Learning, 2021.

16. Pierre K and Pellerin L. Monocarboxylate transporters in the central nervous system: Distribution, regulation and function. *J Neurochem* 2005; 94: 1–14.

17. Killick R, Fearnhead P and Eckley IA. Optimal detection of changepoints with a linear computational cost. *J Am Stat Assoc* 2012; 107: 1590–1598.

18. Truong C, Oudre L and Vayatis N. Selective review of offline change point detection methods. *Signal Process* 2020; 167: 107299.

19. Satopaa V, Albrecht J, Irwin D, et al. Finding a “Kneedle” in a haystack: Detecting knee points in system behavior. In: *31st International Conference on Distributed Computing Systems Workshops*, Minneapolis, MN, USA, 20–24 June 2011, pp.166–171. New York: IEEE.

20. Arvai K. *kneed* [computer program]. Version 0.8.5. Zenodo, 2023. DOI: 10.5281/zenodo.8127224.

21. Kievit R, Frankenhuis WE, Waldorp L, et al. Simpson's paradox in psychological science: A practical guide. *Front Psychol* 2013; 4: 513.

22. Hackett EP, Pinho MC, Harrison CE, et al. Imaging acute metabolic changes in patients with mild traumatic brain injury using hyperpolarized [1-$^{13}$C]pyruvate. *iScience* 2020; 23: 101885.

23. Khellaf A, Khan DZ and Helmy A. Recent advances in traumatic brain injury. *J Neurol* 2019; 266: 2878–2889.

24. Pawlowski M, Ortmann D, Bertero A, et al. Inducible and deterministic forward programming of human pluripotent stem cells into neurons, skeletal myocytes, and oligodendrocytes. *Stem Cell Reports* 2017; 8: 803–812.

25. Tourigny DS, Karim MKA, Echeveste R, et al. Energetic substrate availability regulates synchronous activity in an excitatory neural network. *PLoS One* 2019; 14: e0220937.

26. Baranes K, Hastings N, Rahman S, et al. Transcription factor combinations that define human astrocyte identity encode significant variation of maturity and function. *Glia* 2023; 71: 1870–1889.

27. Watmuff B, Habibollahi F, Desouza C, et al. Drug treatment alters performance in a neural microphysiological system of information processing. *Commun Biol* 2025; 8: 916.

# Titles and legends to figures

**Figure 1. Cerebral microdialysis and the linear lactate–pyruvate framework. Left**: schematic of the intracranial multimodality monitoring bolt and CMD catheter (ICP and $PbtO_2$ probes shown). **Bottom right**: close-up of the CMD catheter tip sampling extracellular fluid. CMD schematic adapted from Khellaf et al.[23] under Creative Commons License (CC BY). **Top–middle right**: simulated CMD data illustrating the proposed framework, generated using representative parameters (cohort medians) from the Cambridge dataset. **Top right**: LPR vs. time for two contiguous segments over ~83 hours, with a hypothetical intervention at hour ~40. Despite the underlying lactate–pyruvate gradient (LPG) decreasing from 29.04 to 17.97 between segments, the observed LPR remains in a similar, overlapping range, illustrating how the conventional LPR time series can obscure true changes in anaerobic activity. **Middle right**: corresponding lactate vs. pyruvate scatter plots with OLS regression lines. The LPG of the first segment (gold triangles; Type $N_b$, LPG = 29.04, $b = -0.55$ mM) reflects higher anaerobic activity than the second segment (teal circles; Type $P_b$, LPG = 17.97, $b = 0.87$ mM). ***b***, linear intercept; **CMD**, cerebral microdialysis; **ICP**, intracranial pressure; **LPG**, lactate–pyruvate gradient; **LPR**, lactate/pyruvate ratio; $\mathbf{N}_{\boldsymbol{b}}$, negative intercept ($b < 0$); **OLS**, ordinary least squares; $\mathbf{P}_{\boldsymbol{b}}$, positive intercept ($b > 0$); $\mathbf{PbtO_2}$, brain tissue partial pressure of oxygen.

**Figure 2. Lactate–pyruvate segmentation results for 12 representative patients.** Each panel (**A**–**L**) shows one patient's segment-level scatter plots with ordinary least squares regression lines. Datapoints are coloured according to their assigned segment, which are temporally distinct and sequential; for any given patient, all datapoints belonging to 'Segment 1' occur before those in 'Segment 2', and so on. Each segment comprises a contiguous block of hourly observations bounded by changepoints. Segments are distinctly coloured in temporal order (orange first, blue second, then green). All segments shown for these patients had positive lactate–pyruvate gradients ($m > 0$).

**Figure 3. Empirical lactate–pyruvate relationships across retained segments.** Left panels display lactate vs. pyruvate (linear space); right panels display LPR vs. pyruvate (hyperbolic space). Thin lines represent individual segment-level ordinary least squares fits; thick lines represent the cohort median trend. Dashed horizontal lines indicate the LPG (*m*) asymptote. (**A**–**B**) Unstratified: all 1,339 retained segments from 506 patients, coloured by intercept type (teal = Type $P_b$, gold = Type $N_b$) with cohort median trend (brown). (**C**–**D**) Type $P_b$ segments: 967 segments, 476 patients. Positive intercepts produce LPR hyperbolae that decrease toward the LPG asymptote as pyruvate increases. (**E**–**F**) Type $N_b$ segments: 372 segments, 276 patients. Negative intercepts produce LPR hyperbolae that increase toward the LPG asymptote as pyruvate increases. (**G**–**H**) Combined overlay comparing Type $P_b$ and Type $N_b$ cohort median trends. ***b***, linear intercept; **LPG (*m*)**, lactate–pyruvate gradient; **LPR**, lactate/pyruvate ratio; $\mathbf{N}_b$, negative intercept ($b < 0$); $\mathbf{P}_b$, positive intercept ($b > 0$).

**Figure 4. LPR deviation from LPG in a representative patient.** (**A**) Lactate vs. pyruvate for the patient's two retained segments (linear fits overlaid). (**B**) The initial Type $N_b$ segment (gold triangles; LPG = 28.76, $b$ = -0.74 mM, $r$ = 0.92). (**C**) The subsequent Type $P_b$ segment (teal circles; LPG = 10.67, $b$ = 0.72 mM, $r$ = 0.74). (**D**) LPR (y-axis) plotted against time since injury (hours) for both segments. Dashed horizontal lines indicate the LPG for each segment; shaded regions highlight the deviation of the observed LPR from LPG. In the Type $N_b$ segment, LPR (range: 5.9–27.7) consistently underestimated the LPG, while in the Type $P_b$ segment, LPR (range: 14.8–28.5) consistently exceeded the LPG. Despite the LPG decreasing from 28.76 to 10.67 between segments, the observed LPR remained in a similar range due to the opposing effects of the intercept, illustrating how LPR can obscure true changes in anaerobic activity. ***b***, linear intercept; **LPG**, lactate–pyruvate gradient; **LPR**, lactate/pyruvate ratio; $\mathbf{N}_b$, negative intercept ($b < 0$); $\mathbf{P}_b$, positive intercept ($b > 0$); ***r***, Pearson correlation coefficient.

**Figure 5. Substrate delivery and LPR fidelity.** (**A**) Within-segment LPR vs. glucose for Type $P_b$ segments (954 segments, 474 patients): LPR decreases with increasing glucose ($\beta_{1/\mathrm{glucose}}$ = 1.21 mM, $p < 0.001$). (**B**) Within-segment LPR vs. glucose for Type $N_b$ segments (367 segments, 272 patients): LPR increases with increasing glucose ($\beta_{1/\mathrm{glucose}}$ = -0.48 mM, $p < 0.001$). (**A**–**B**) Within-segment models use 1/Glucose as predictor to capture the hyperbolic LPR–glucose relationship (LPR = $m + b/\mathrm{P}$); predictions are plotted on the glucose axis; nested random intercepts (segment within patient); $p$-values are Benjamini-Hochberg corrected (n = 2). (**C**) Segment-level median LPR vs. LPG ($\beta$ = 0.255, $p < 0.001$); dashed black line indicates the theoretical $\beta$ = 1 relationship expected if LPR perfectly tracked LPG. (**D**) Segment-level LPG (blue; $\beta$ = -1.75 $\mathrm{mM}^{-1}$, $p < 0.001$) and median LPR (orange; $\beta$ = -2.41 $\mathrm{mM}^{-1}$, $p < 0.001$) vs. median glucose. (**C**–**D**) Patient random intercepts, weighted by segment size. (**A**–**D**) Cambridge cohort; lines represent LME fixed effects; shaded regions represent 95% CIs; sample: 1,321 segments, 504 patients. (**E**–**H**) Corresponding analyses for the Uppsala cohort (215 segments, 81 patients). (**E**) Type $P_b$ (132 segments, 72 patients; $\beta_{1/\mathrm{glucose}}$ = 1.72 mM, $p < 0.001$). (**F**) Type $N_b$ (83 segments, 61 patients; $\beta_{1/\mathrm{glucose}}$ = -0.11 mM, $p$ = 0.090). (**G**) Median LPR vs. LPG ($\beta$ = 0.140, $p$ = 0.005). (**H**) LPG vs. glucose ($\beta$ = -1.20 $\mathrm{mM}^{-1}$, $p$ = 0.022) and median LPR vs. glucose ($\beta$ = -3.95 $\mathrm{mM}^{-1}$, $p < 0.001$). ***b***, linear intercept; **CI**, confidence interval; **LME**, linear mixed-effects model; **LPG (*m*)**, lactate–pyruvate gradient; **LPR**, lactate/pyruvate ratio; $\mathbf{N}_b$, negative intercept ($\boldsymbol{b} < 0$); $\mathbf{P}_b$, positive intercept ($\boldsymbol{b} > 0$).

**Figure 6. Estimated marginal means of linear lactate–pyruvate parameters by 6-month functional outcome.** (**A**) Segment median LPR, (**B**) segment LPG, and (**C**) segment *b* across four GOSE categories: Death (GOSE 1; 87 patients, 217 segments), Severe disability (GOSE 2–4; 117 patients, 310 segments), Moderate disability (GOSE 5–6; 160 patients, 423 segments), and Good recovery (GOSE 7–8; 112 patients, 305 segments). Points represent EMMs from weighted LME models (weights = segment size) with patient random intercepts; error bars represent 95% CIs. Linear contrasts tested for monotonic trends across outcome categories: (**A**) LPR, $\beta = -10.27$ ($p = 0.030$); (**B**) LPG, $\beta = -5.59$ ($p = 0.039$); (**C**) *b*, $\beta = 0.10$ ($p = 0.806$). (**A**–**C**) Cambridge cohort. (**D**–**F**) Corresponding analyses for the Uppsala cohort: Death (6 patients, 20 segments), Severe (29 patients, 74 segments), Moderate (16 patients, 41 segments), Good (18 patients, 52 segments). Linear contrasts: (**D**) LPR, $\beta = -1.59$ ($p = 0.470$); (**E**) LPG, $\beta = 6.51$ ($p = 0.784$); (**F**) *b*, $\beta = -1.17$ ($p = 0.546$). ***b***, linear intercept; **CI**, confidence interval; **EMM**, estimated marginal mean; **GOSE**, Glasgow Outcome Scale-Extended; **LME**, linear mixed-effects model; **LPG**, lactate–pyruvate gradient; **LPR**, lactate/pyruvate ratio.

**Figure 7. *In vitro* lactate–pyruvate relationship under rotenone-induced mitochondrial dysfunction.** Lactate vs. pyruvate for control (blue; n = 34; LPG = 14.2, $b = 0.026$ mM, $R^2 = 0.66$) and rotenone-treated (red; n = 34; LPG = 32.9, $b = -0.107$ mM, $R^2 = 0.80$) hiPSC-derived neurons. ANCOVA confirmed a significant shift in LPG ($p < 0.001$) and *b* ($p < 0.001$; residual df = 64). Points represent individual measurements from four independent experiments, each condition having $\geq 2$ technical replicates, pooled across timepoints between 1 and 24 hours. Lines represent OLS regression fits; shaded regions represent 95% CIs. **ANCOVA**, analysis of covariance; ***b***, linear intercept; **CI**, confidence interval; **hiPSC**, human induced pluripotent stem cell; **LPG**, lactate–pyruvate gradient; **OLS**, ordinary least squares.